

DESIGN

THINKING AND

INNOVATION

BATCH-11

Vigilance Drive

-A Sleep Sensing Glasses-

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Abstract:

Drowsy driving is a significant problem that leads to accidents and fatalities worldwide. Existing solutions to detect drowsiness have limitations and cannot be used in small vehicles like bikes.

Our proposed system, "Sleep alert glasses", uses an Eye Blink sensor and Arduino to monitor driver's eye movements and trigger a warning using vibrations and a loud buzzer if the driver is dozing off . The system also uses RF Transmitter and Receiver modules for data processing.

By detecting the onset of drowsiness early, Sleep alert glasses have the potential to significantly reduce accidents caused by driver fatigue and make roads safer for everyone.

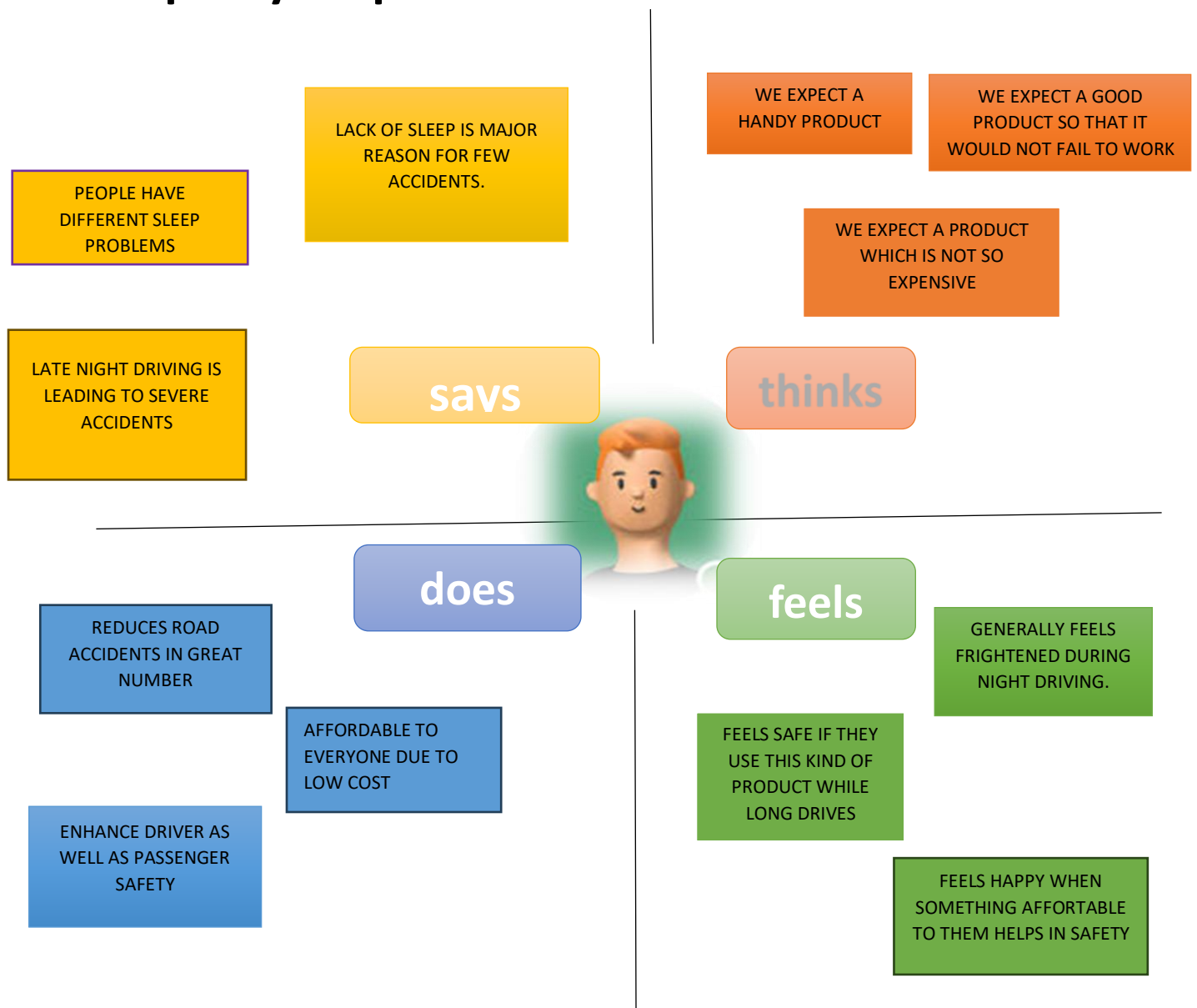
Keywords: : Eye Blink sensor, Arduino ,Vibration sensor, Buzzer alert, Sleep Alert Glasses

Introduction:

Drowsy driving is a global issue that poses a significant threat to road safety. According to the National Sleep Foundation, a large percentage of adults have driven while feeling drowsy or even fallen asleep at the wheel. This has resulted in a significant number of accidents, injuries, and deaths, with approximately 100,000 accidents each year attributed to drowsy driving in the United States alone. Drowsy driving is not limited to the US, as it is also a problem around the world. In Australia, it is responsible for around 20% of all fatal crashes, while in Europe it accounts for up to 30% of all road accidents. In Japan, it is the most common cause of fatal accidents on highways. To combat this issue, various measures have been introduced, including rest areas, warning signs, and education campaigns. It is essential to detect drowsiness to prevent accidents, and several indirect methods such as monitoring changes in a driver's behavior and physiological responses can be used. Advanced

technology like sleep alert glasses, which use infrared sensors to detect eye movement and alert the driver with a sound or vibration if signs of drowsiness are detected, is also being developed. Preventing drowsy driving requires a combination of technology and education. By raising awareness about the hazards of driving while tired and encouraging drivers to prioritize getting enough sleep and taking breaks, we can work towards a safer and more responsible driving culture around the world.

Empathy map:



USERS

SAYS:

1. PEOPLE HAVE DIFFERENT SLEEP PROBLEMS
2. LACK OF SLEEP IS MAJOR REASON FOR FEW ACCIDENTS.
3. LATE NIGHT DRIVING IS LEADING TO SEVERE ACCIDENTS

Thinks:

1. WE EXPECT A HANDY PRODUCT
2. WE EXPECT A GOOD PRODUCT SO THAT IT WOULD NOT FAIL TO WORK
3. WE EXPECT A PRODUCT WHICH IS NOT SO EXPENSIVE

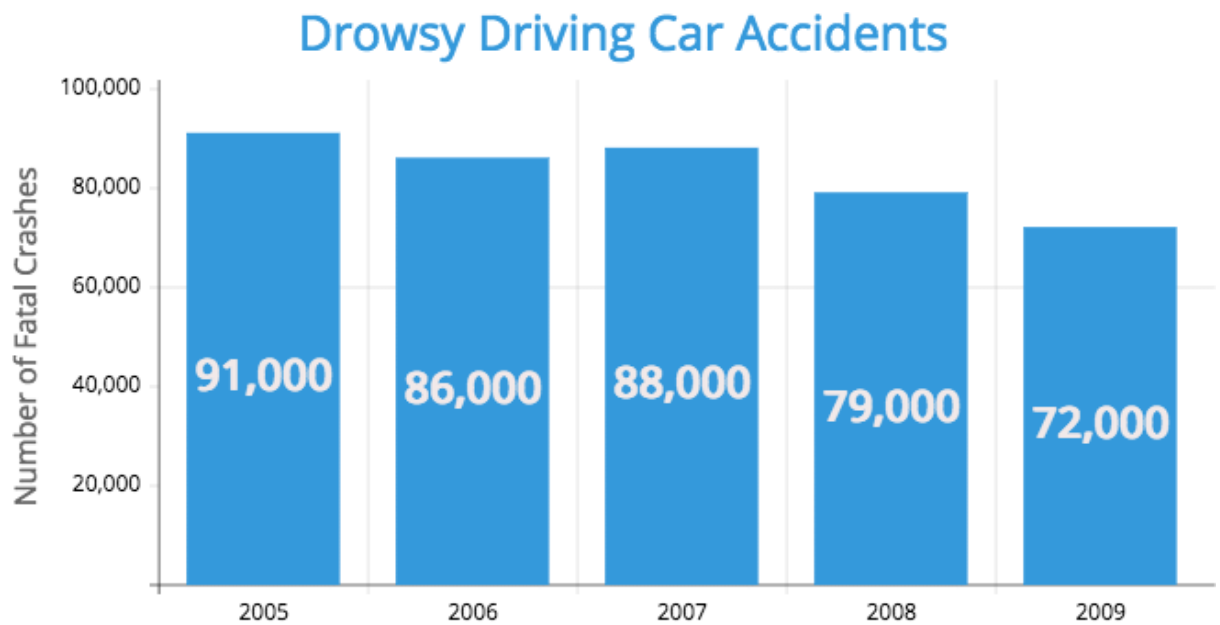
DOES:

1. REDUCES ROAD ACCIDENTS IN GREAT NUMBER
2. AFFORDABLE TO EVERYONE DUE TO LOW COST
3. DRIVER AS WELL AS PASSENGER SAFETY

Feels:

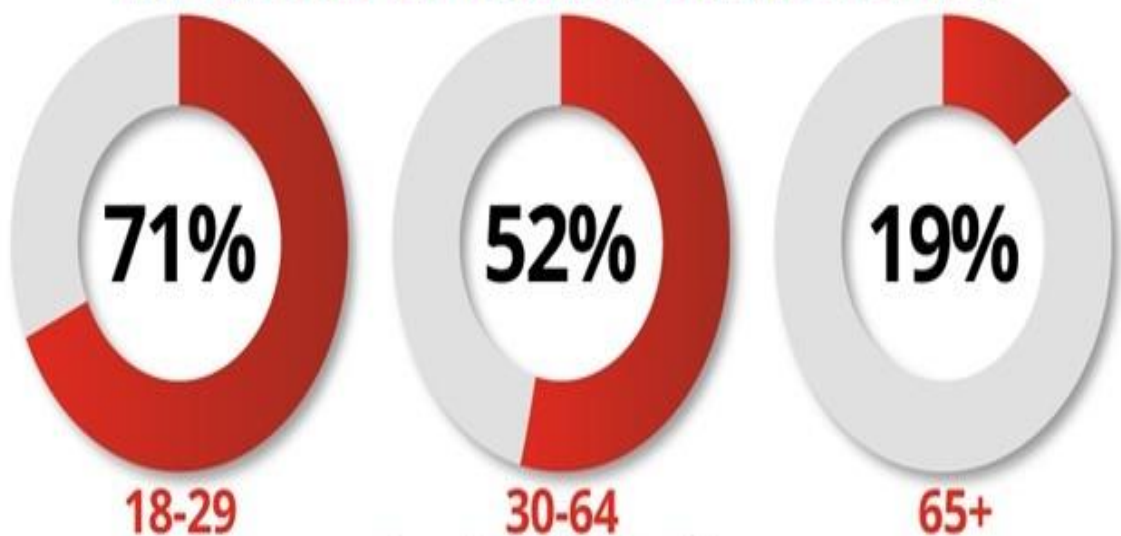
1. GENERALLY FEELS FRIGHTENED DURING NIGHT DRIVING
2. FEELS SAFE IF THEY USE THIS KIND OF PRODUCT WHILE LONG DRIVES
3. FEELS HAPPY WHEN SOMETHING AFFORDABLE TO THEM HELPS IN SAFETY

PREDICTIVE ANALYSIS:



Source: NHTSA - Drowsy Driving

LIKELIHOOD OF DROWSY DRIVING BY AGE



Source: National Sleep Foundation

SURVEY ANALYSIS

How often do you experience difficulty falling asleep?

52 responses

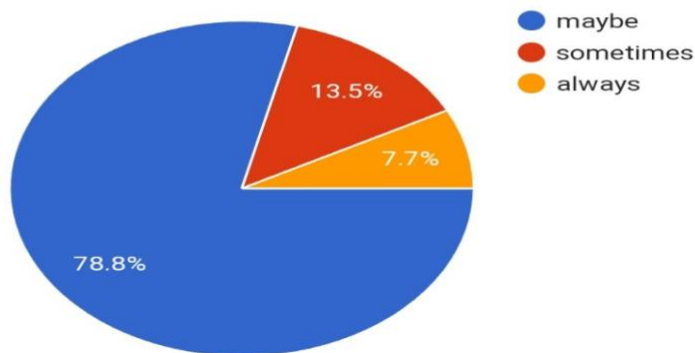


Fig1

- From this (fig1) we have understood that many of people are facing difficulty in sleeping during nights or experiencing difficulty falling asleep.

2)

Would you be concerned about privacy when using sleep sensing glasses?

51 responses

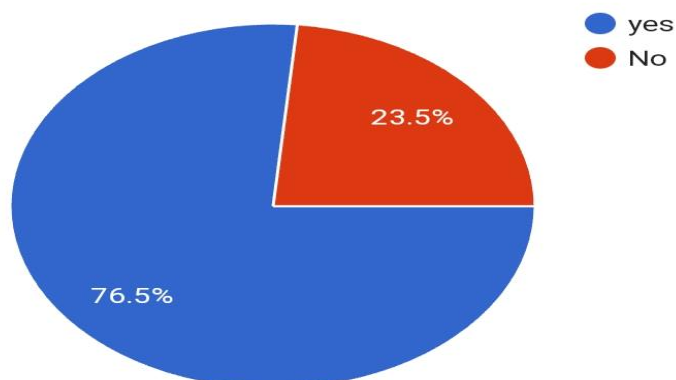


Fig 2

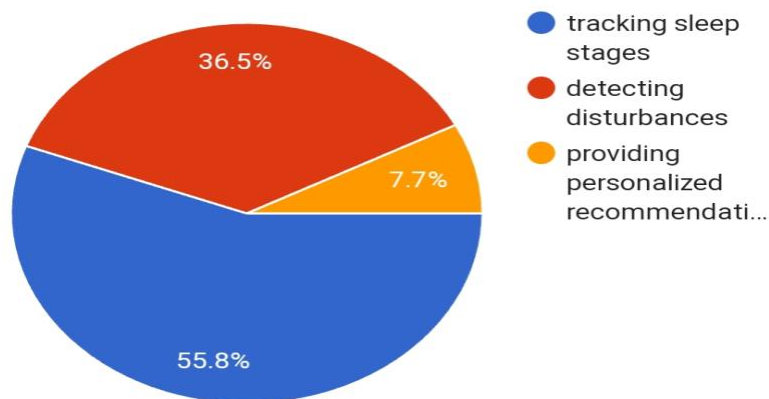
- from fig 2 we understood that many people are concerned about sleep sensing glass

3)

What features would you most value in sleep sensing glasses?

52 responses

 Copy Fig 3

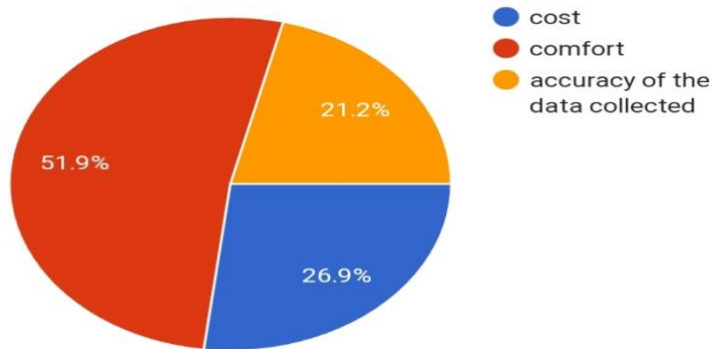


- From this (fig 3) we have concluded that many of people chooses health tracking and Integration as the most value in sleep sensing glasses

What factors would influence your decision to try or adopt sleep sensing glasses?



52 responses

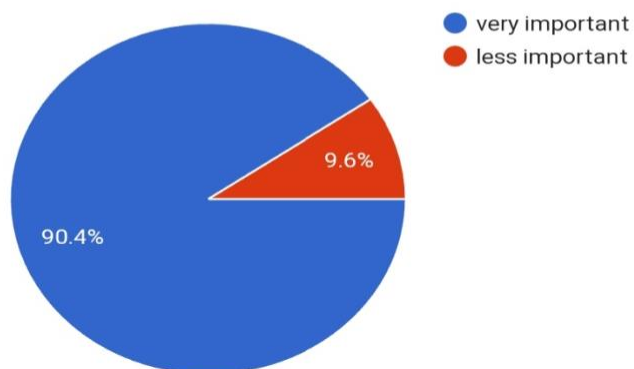


From this fig4 we have understood that many people suggest for comfort and accuracy in using the sleep sensing glasses

How important is it for sleep sensing glasses to have a user-friendly interface or app for viewing sleep data?



52 responses



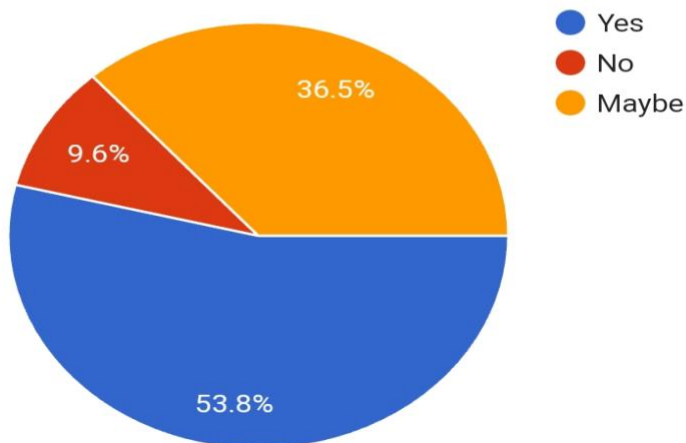
- from this we have understood many people feels sleep sensing glasses as user friendly.

6)

Do you think sleep sensing glasses could be beneficial for improving your overall sleep quality?

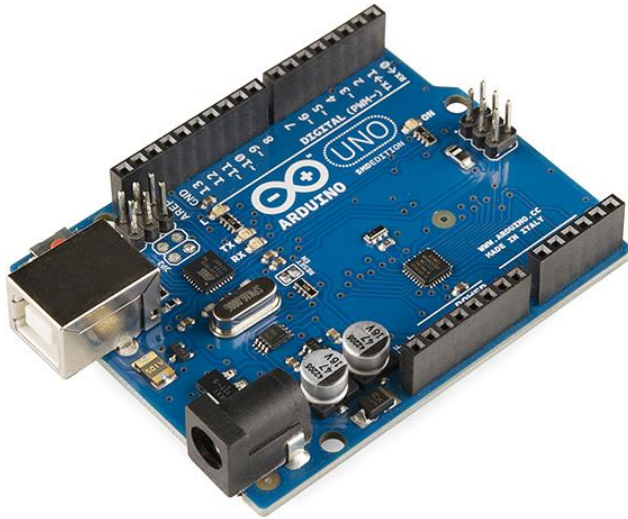


52 responses



- From this we can conclude that many people feels that sleep sensing glasses would be beneficial.

DEFINITION PHASE



AURDINO DATA SHEET

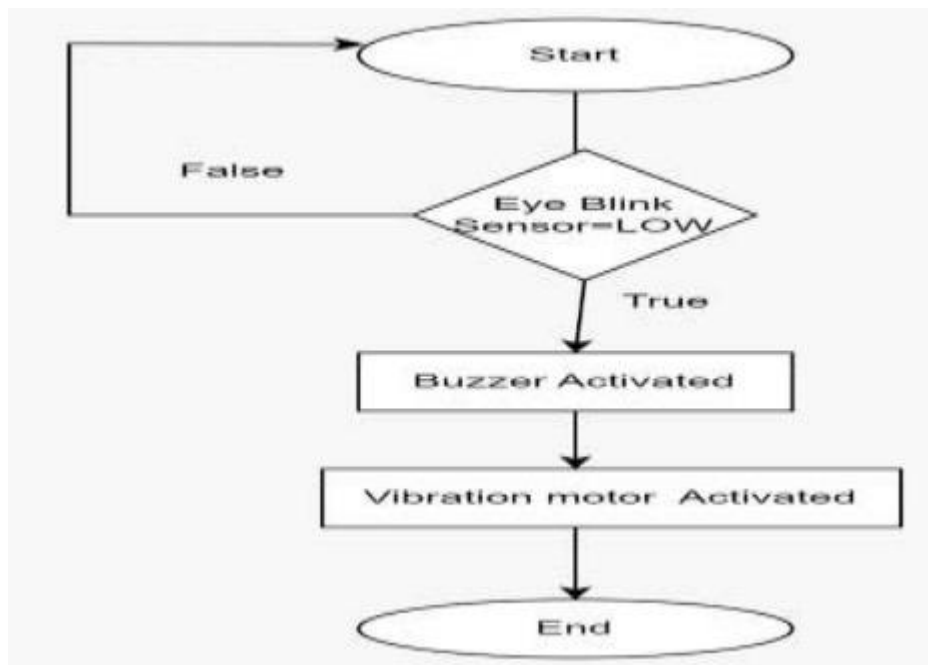


IR SENSOR



BUZZER

The drowsiness detection system that analyzes the driver's respiratory signals using a piezoelectric sensor. Their system detects changes in the driver's respiratory pattern and uses these changes to determine the level of drowsiness as proposed by , a hybrid approach is suggested, which combines wearable sensors and machine learning algorithms for drowsiness detection. Their system uses both physiological signals and motion data obtained from wearable sensors to detect drowsiness in drivers. [3] proposes a system that uses an Arduino board and sensors to detect drowsiness in bikers wearing helmets. The system uses an eye blink detection algorithm and a head tilt detection algorithm to detect drowsiness in bikers. [4] proposes a deep learning-based approach that uses a condition-adaptive representation learning framework for drowsiness detection in drivers. warns the driver in real-time. In, a deep learning-based approach is proposed that uses a IR SENSOR to detect drowsiness in drivers. Their system uses both physiological signals and driving behavior data obtained from sensors to detect drowsiness in drivers. proposes a system that uses eye aspect ratio (EAR) to detect drowsiness in drivers and a smart alarm system to alert the driver. Finally, proposes a drowsiness detection system that takes into account individual differences in physiological signals to detect drowsiness in drivers. Their system achieves high accuracy in detecting drowsiness and considers individual differences in physiological signals to improve the system's performance.



CASE STUDIES:

PERSONAL PROBLEM:

BY drowsiness accidents drivers may affect large permanent injuries .it effects to person life long.

FINANCIAL PROBLEM :

A victim shares the financialhardship faced after being hit by a driver fatuge, with the medical bills and they lost income adding stress to their trauma.

FAMILY BURDEN

Parents recount the day they lost their child SLEEP WHILE DRIVING and tragedy that tore their family apart irreparably.

EFFECT TO OTHERS:

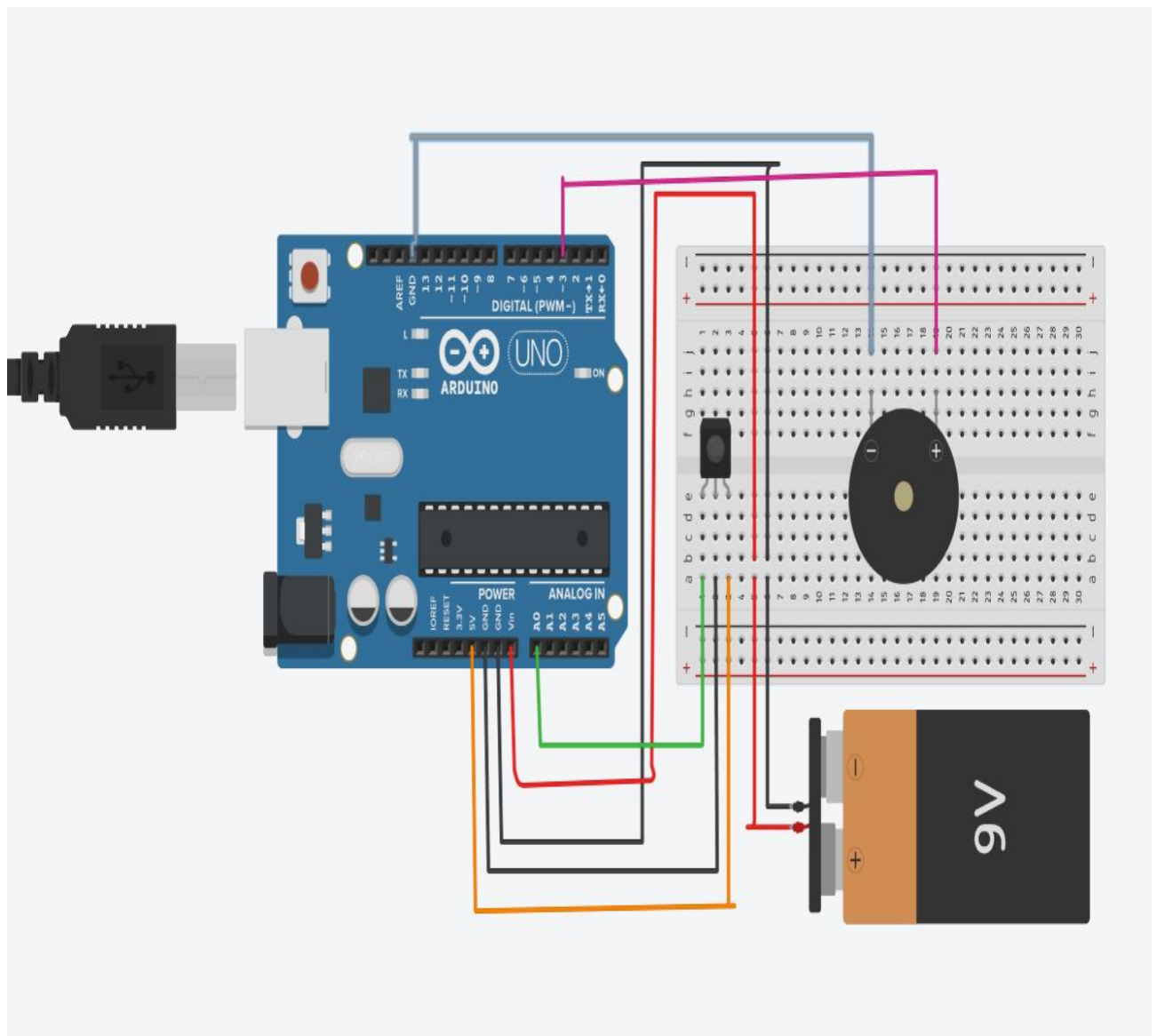
Due to driving a vehicle while sleep it causes affect to other drivers in the same road or adjacent lane.their life is also in risk.

PROPOSED SYSTEM

The proposed system is a wearable glasses named – Sleep Alert glasses. These glasses monitor the eye movements of the driver using an Eye Blink sensor. If the driver is found to be acting drowsy, i.e; closing their eyes for more than two seconds. The system will detect that behavior and trigger an alert using a loud buzzer and sending vibrations. Light-weight components such as Arduino nano, 3V buzzer, vibrator motor, are used for the creation of the system so that the drivers can easily wear them for a long time without any inconvenience. This device will be more efficient as it is user-friendly across all vehicles and its performance does not get affected by driving or weather conditions. The Sleep alert glasses are designed primarily as a tool for improving driver safety, especially for those who are at high risk of drowsy driving, such as long-haul truck drivers, nighttime drivers, and those driving long distances alone or suffering from sleep deprivation. However, the device can also be used in other contexts to prevent drowsiness and promote alertness. The design methodology used to create the Sleep alert glasses involves an iterative research and analysis cycle. During the research stage, a range of concepts is generated through brainstorming sessions, market research, and user feedback. These concepts are

then evaluated against a set of design requirements and constraints, such as the need for the glasses to be lightweight, comfortable, and easy to use. During the analysis stage, the most promising concepts are selected for further evaluation. This involves testing the glasses under different conditions to ensure they meet the design requirements and user needs. By continually evaluating and improving the design, the Sleep alert glasses can provide an effective solution for preventing drowsy driving and promoting safety on the road.

PROTOTYPE



RESULT ANALYSIS

The results show that if the eyes remain closed for one second or less, the buzzer will remain off. If the eyes are closed for two seconds, the buzzer will ring. If the eyes are open, the buzzer will remain off. The above implemented system works well in all lighting conditions.

Basis	Condition	Buzzer status
Duration	Eye Blink	remains off
	Eyes closed for one second	remains off
	Eyes closed for two seconds	rings
	Eyes open	remains off

If the lighting is bright, normal, or dark, and the person is drowsy, the buzzer will ring in all cases

Basis	Condition	Buzzer status when drowsy
Lighting	Bright light	Rings
	Normal light	Rings
	Dark	Rings



CONCLUSION

The Driver Drowsiness Detection and Alerting System is a critical technology designed to prevent accidents caused by driver fatigue. The system, based on Arduino platform, uses an eyeblink sensor to detect drowsiness and alert drivers in real-time. The system is easy to build, cost-effective, and highly efficient, making it an ideal solution for a wide range of vehicles. To enhance the system, machine learning algorithms can be incorporated to improve the accuracy of drowsiness detection, and the system can be integrated with other safety features, such as collision avoidance systems. Wireless communication technology, such as Wi-Fi or Bluetooth, can be used to communicate with other devices, providing additional alerts or notifications.

*****GOOD BED - - - -SAFE RIDE*****